

Mrs Cooper
Mr Feng
Mrs Kerr
Mrs Millar
Mrs Squires

Ms Davis
Mrs Greenberg
Mrs Kydd
Ms Smith
Mrs Ward

Name:

Teacher:.....



Pymble Ladies' College

YEAR 11 MATHEMATICS ADVANCED

ASSESSMENT TASK 2

Term 2

2022

General Instructions

- Working time 60 minutes
- Write using black or blue pen.
- Erasable pens are not to be used.
- Approved calculators may be used.
- Diagrams are not drawn to scale.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or untidy work.
- Reference sheets are provided.

Mark	/45
Rank	/
Highest Mark	/45

Section I

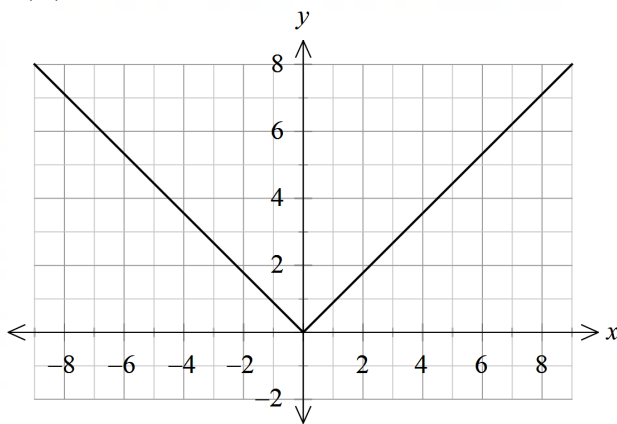
Multiple Choice.

(Each question in this section is worth 1 mark)

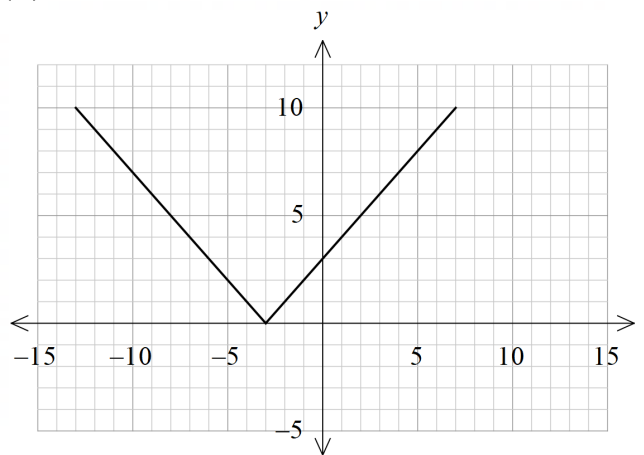
Answer the Multiple Choice questions by placing a circle around your answer.

1 Which of these is the graph of $y = |x + 3|$?

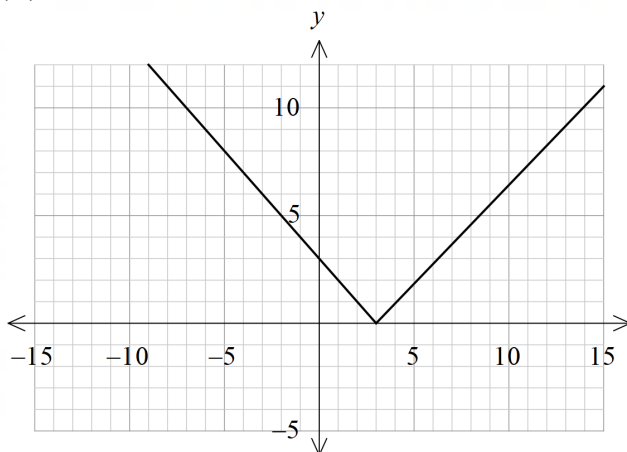
(A)



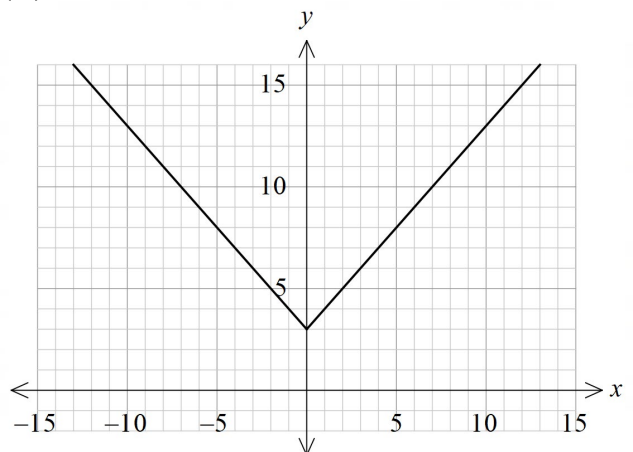
(B)



(C)

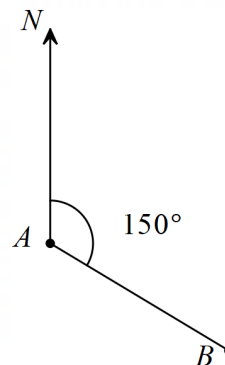


(D)



2 A plane flies on a bearing of 150° from A to B.
What is the bearing from B to A?

- (A) 30°
- (B) 60°
- (C) 300°
- (D) 330°



3 What is the correct simplification of the expression $\frac{1}{\sec 30^\circ - \tan 30^\circ}$?

(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{2}$

(D) $\frac{\sqrt{3}}{2}$

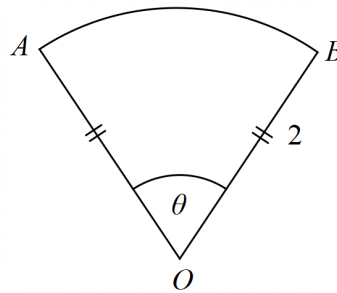
4 The diagram shows a sector OAB of a circle with centre O and radius 2. The angle AOB is θ radians and the perimeter of the sector is 6. What is the correct value for θ ?

(A) 1

(B) 2

(C) $\sqrt{3}$

(D) 3



5 What is the correct derivative of $f(x) = x^2 + 6x + 9$?

(A) $f'(x) = 2x + 6 + 9$

(B) $f'(x) = x^2 + 6$

(C) $f'(x) = 2x + 6$

(D) $f(x) = 2x$

6 What is the correct derivative of $y = (x^2 + 5)^3$?

(A) $6(x^2 + 5)^2$

(B) $6x(x^2 + 5)^2$

(C) $(2x + 5)^3$

(D) $3(2x + 5)^2$

7 What is the exact value of $\cos\left(\frac{3\pi}{4}\right)$?

(A) $-\frac{1}{\sqrt{2}}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{2}$

(D) $-\frac{1}{2}$

8 The derivative of $y = 2x^{\frac{1}{2}}$ is which of the following?

(A) $-\frac{2}{\sqrt{x}}$

(B) $-\frac{1}{\sqrt{x}}$

(C) $\frac{1}{\sqrt{x^3}}$

(D) $\frac{1}{\sqrt{x}}$

9 The equation $x^2 + y^2 - 6x + 6 = 0$ represents a circle. What are the coordinates of the centre of the circle?

(A) $(3, -3)$

(B) $(-3, -3)$

(C) $(3, 0)$

(D) $(-6, 0)$

10 What is the exact value of $\tan(-150^\circ)$?

(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) $-\sqrt{3}$

(D) $-\frac{1}{\sqrt{3}}$

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Section II

Name.....

Attempt all Questions.

Your responses should include relevant mathematical reasoning and/or calculations.

Answer in the spaces provided.

Question 11 (2 marks)

State the domain and range of $y = 2 + |x - 3|$.

2

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Question 12 (3 marks)

(a) Sketch the graph of $y = \sin x$ for $0 < x < 2\pi$.

2



(b) How many solutions are there to $\sin x = -0.4$ for $0 < x < 2\pi$?

1

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Question 13 (1 mark)

Convert $\frac{4\pi}{7}$ to degrees, giving your answer to the nearest minute.

1

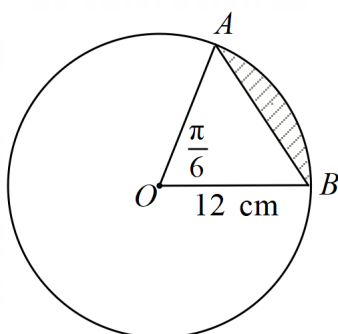
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Question 14 (3 marks)

The diagram shows a circle with radius 12 cm, centre O .



Given $\angle AOB = \frac{\pi}{6}$:

- (a) Find the area of the sector AOB .

1

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- (b) Find the exact area of the shaded segment.

2

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Question 15 (2 marks)

Simplify $\cot \theta \sec \theta$.

2

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Question 16 (3 marks)

Given that $\cos \theta = \frac{\sqrt{5}}{3}$ and $\tan \theta < 0$, find the exact value of $\sin \theta \sec \theta$.

3

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Question 17 (8 marks)

Differentiate the following.

(a) $y = x^2 + x$ **1**

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(b) $y = 7x^4 + \frac{6}{x} - 9$ **2**

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(c) $f(x) = \frac{x^2}{4x+1}$ **2**

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(d) $y = x^2(x-1)^3$ **3**

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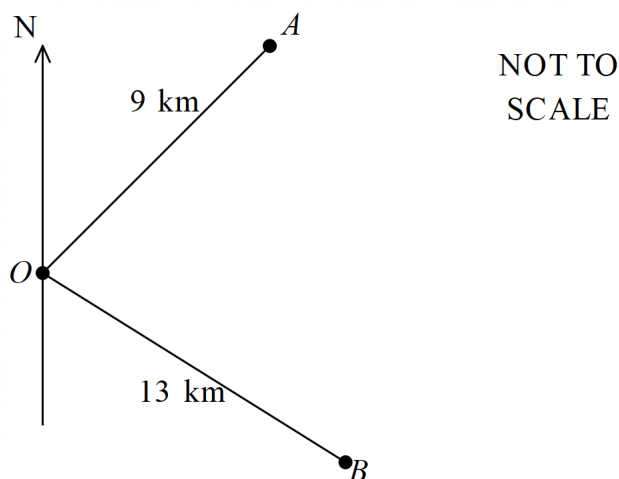
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Question 18 (4 marks)

Alex and Bobby and a group of students were camping at a site. Alex walked with some students on a bearing of 037° for 9 km to location A . Bobby, with the rest of the students, walked on a bearing of 102° for 13 km to location B .



- (a) Show that the angle AOB is 65° . 1

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- (b) Find the distance AB , in km, correct to 2 decimal places. 1

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- (c) Find the bearing of Bobby's group from Alex's group. 2
Give your answer correct to the nearest degree.

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Question 19 (4 marks)

Solve the following equations:

(a) $\cos \theta = -\frac{1}{\sqrt{2}}, \quad 0 < \theta < 360^\circ$

1

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(b) $2 \sin^2 \theta - 3 \sin \theta = -1, \quad 0 < \theta < 2\pi$

3

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Question 20 (2 marks)

Find the values of α and x such that:

2

$$\alpha \sin x = 5 \sin 20^\circ + 3 \cos 70^\circ.$$

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Question 21 (3 marks)

Prove that $\frac{1}{\cos \theta} + \cos \theta = (2 - \sin^2 \theta) \sec^2 \theta$.

3

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End of Assessment

Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a full page of white paper designed for handwriting practice. It features approximately 20 evenly spaced horizontal dotted lines running across the width of the page. There are no margins, text, or other markings present.

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Section I

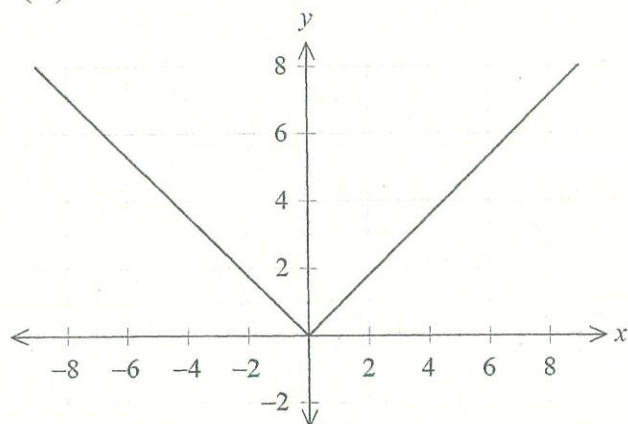
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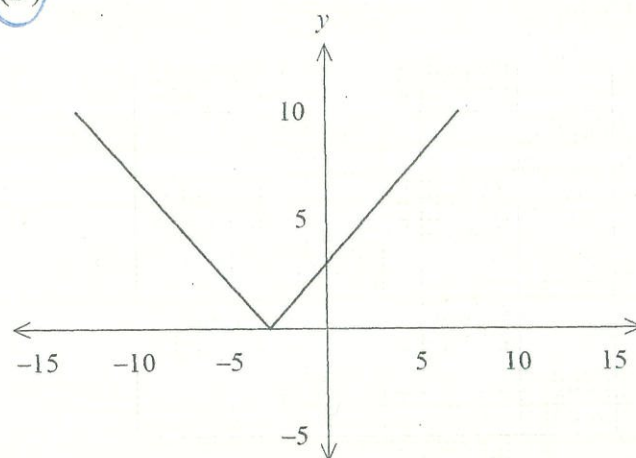
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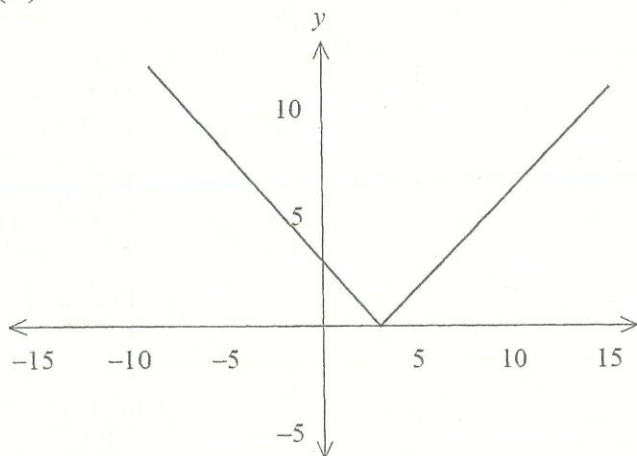
(A)



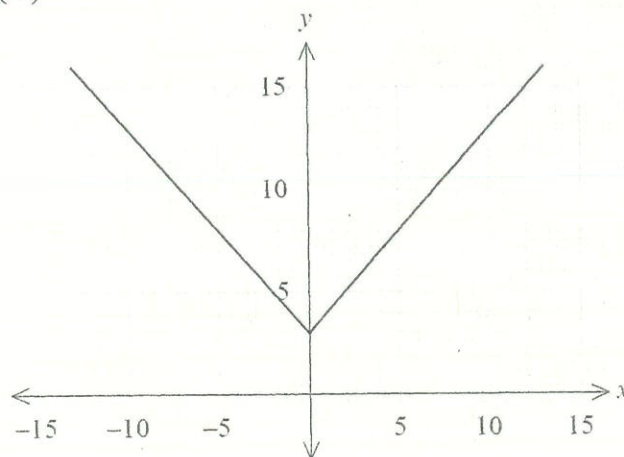
(B)



(C)



(D)



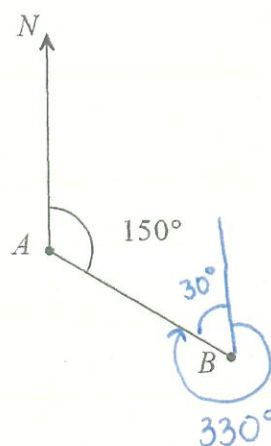
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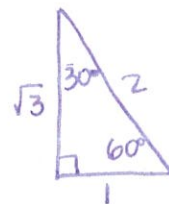
(C) $\frac{1}{2}$

(D) $\frac{\sqrt{3}}{2}$

$$= \frac{1}{\frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}}}$$

$$= \frac{1}{\frac{1}{\sqrt{3}}}$$

$$= \sqrt{3}$$



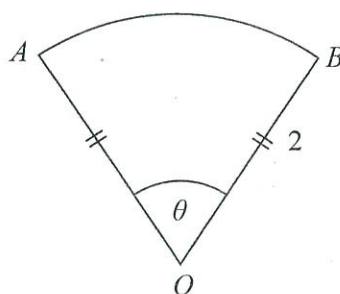
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(A) 1

(B) 2

(C) $\sqrt{3}$

(D) 3



$$P = 6$$

$$6 = 2(2) + 2\theta$$

$$6 = 4 + 2\theta$$

$$2 = 2\theta$$

$$1 = \theta$$

- 5 What is the correct derivative of $f(x) = x^2 + 6x + 9$?

(A) $f'(x) = 2x + 6 + 9$

$$f'(x) = 2x + 6$$

(B) $f'(x) = x^2 + 6$

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(A) $6(x^2 + 5)^2$

(B) $6x(x^2 + 5)^2$

(C) $(2x + 5)^3$

(D) $3(2x + 5)^2$

$$y' = 3(x^2 + 5)^2 (2x)$$

$$= 6x(x^2 + 5)^2$$

7 What is the exact value of $\cos\left(\frac{3\pi}{4}\right)$?

(A) $-\frac{1}{\sqrt{2}}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{2}$

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8 The derivative of $y = 2x^{\frac{1}{2}}$ is which of the following?

(A) $-\frac{2}{\sqrt{x}}$

(B) $-\frac{1}{\sqrt{x}}$

(C) $\frac{1}{\sqrt{x^3}}$

(D) $\frac{1}{\sqrt{x}}$

$$y' = x^{-1/2}$$

$$= \frac{1}{\sqrt{x}}$$

9 The equation $x^2 + y^2 - 6x + 6 = 0$ represents a circle. What are the coordinates of the centre of the circle?

(A) $(3, -3)$

(B) $(-3, -3)$

(C) $(3, 0)$

(D) $(-6, 0)$

$$x^2 - 6x + 9 + y^2 = -6 + 9$$

$$(x-3)^2 + y^2 = 3$$

$\therefore$ centre at $(3, 0)$

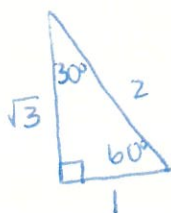
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(A) $\sqrt{3}$

(B) $\frac{1}{\sqrt{3}}$

(C) $-\sqrt{3}$

(D) $-\frac{1}{\sqrt{3}}$



$$\tan 30 = \frac{1}{\sqrt{3}}$$

$$\tan(-150) = \tan 30$$

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Section II

Name.....

Attempt all Questions.

Your responses should include relevant mathematical reasoning and/or calculations.

Answer in the spaces provided.

Question 11 (2 marks)

State the domain and range of $y = 2 + |x - 3|$.

2

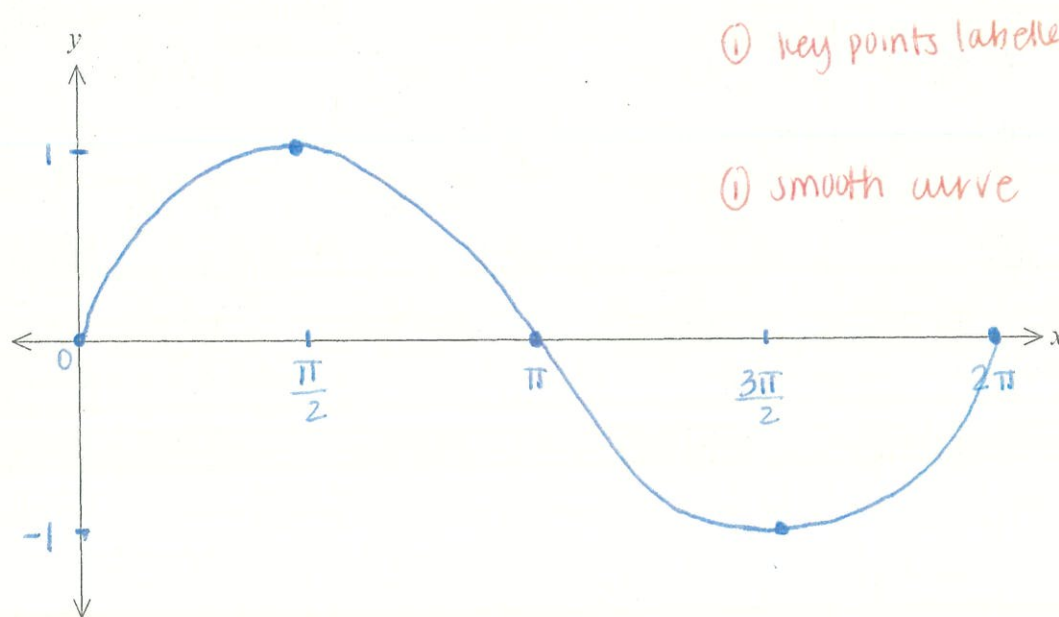
Domain: $x: x \in \mathbb{R}$ ①

Range: $y: y \in [2, \infty]$ ①

Question 12 (3 marks)

(a) Sketch the graph of $y = \sin x$ for $0 < x < 2\pi$.

2



(b) How many solutions are there to $\sin x = -0.4$ for $0 < x < 2\pi$?

1

two solutions ①

Question 13 (1 mark)

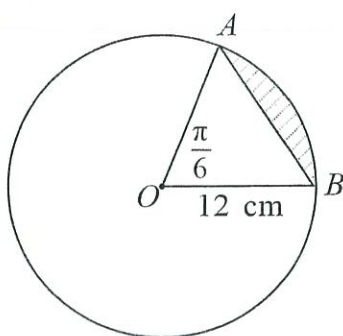
Convert $\frac{4\pi}{7}$ to degrees, giving your answer to the nearest minute.

1

$$\frac{4\pi}{7} \times \frac{180}{\pi} = 102^{\circ}51' \quad (1)$$

Question 14 (3 marks)

The diagram shows a circle with radius 12 cm, centre O .



Given $\angle AOB = \frac{\pi}{6}$:

- (a) Find the area of the sector AOB .

1

$$\begin{aligned} A &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} (12)^2 \left(\frac{\pi}{6} \right) \\ &= 12\pi \text{ cm}^2 \quad (1) \end{aligned}$$

- (b) Find the exact area of the shaded segment.

2

$$\begin{aligned} A_{\text{segment}} &= A_{\text{sector}} - A_{\text{triangle}} \\ &= 12\pi - \frac{1}{2} \times 12 \times 12 \times \sin \frac{\pi}{6} \\ &= (12\pi - 36) \text{ cm}^2 \end{aligned}$$

(1) • Sect - Δ
• correct Δ area formula.
• CPFA - Δ

(2) correct soln

Question 15 (2 marks)

Simplify $\cot \theta \sec \theta$.

2

$$\begin{aligned}\cot \theta \sec \theta &= \frac{\cos \theta}{\sin \theta} \times \frac{1}{\cos \theta} \\ &= \frac{1}{\sin \theta} \\ &= \operatorname{cosec} \theta\end{aligned}$$

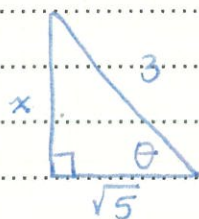
① uses one identity
 $\cot \theta = \frac{\cos \theta}{\sin \theta}$ or $\sec \theta = \frac{1}{\cos \theta}$

② correct solution

Question 16 (3 marks)

Given that $\cos \theta = \frac{\sqrt{5}}{3}$ and $\tan \theta < 0$, find the exact value of $\sin \theta \sec \theta$.

3



$$\begin{aligned}x &= \sqrt{3^2 - (\sqrt{5})^2} \\ &= 2\end{aligned}$$

one is positive & one is negative

$$\sin \theta \sec \theta = -\frac{2}{3} \times \frac{3}{\sqrt{5}}$$

① values of $\sin \theta$ and $\sec \theta$

$$= -\frac{2}{\sqrt{5}}$$

① negative

① • correct Δ
• as $\tan \theta < 0$ Q4

② • $\sin \theta \sec \theta$ using CFPE
• neg sine value.

③ correct soln.

Question 17 (8 marks)

Differentiate the following.

(a) $y = x^2 + x$

1

$y' = 2x + 1$ (1)

(b) $y = 7x^4 + \frac{6}{x} - 9$

2

$= 7x^4 + 6x^{-1} - 9$

$y' = 28x^3 - 6x^{-2}$ (1) (1) OR $y' = 28x^3 - \frac{6}{x^2}$

(c) $f(x) = \frac{x^2}{4x+1} = \frac{u}{v}$ $f'(x) = \frac{vu' - uv'}{v^2}$

2

$f'(x) = \frac{(4x+1)(2x) - (x^2)(4)}{(4x+1)^2}$

(1) correct diff one term

(2) correct solution

$= \frac{8x^2 + 2x - 4x^2}{(4x+1)^2} = \frac{4x^2 + 2x}{(4x+1)^2}$

(d) $y = x^2(x-1)^3 = uv$ $y' = uu' + uv'$

3

$y' = (x-1)^3(2x) + (x^2)(3(x-1)^2)$

(1) correct diff one term.

$= 2x(x-1)^3 + 3x^2(x-1)^2$

OR $= x(x-1)^2(2(x-1) + 3x)$

} not necessary for marking

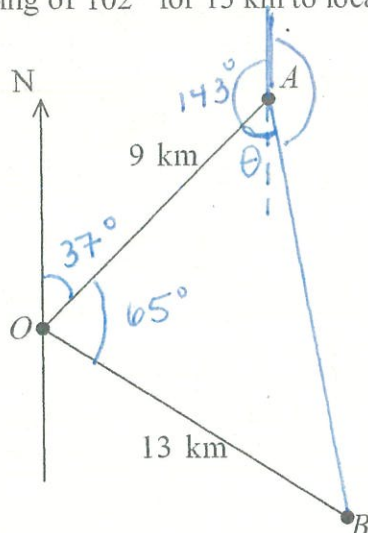
$= x(x-1)^2(5x-2)$

(2) corr diff 2 terms

(3) soln

Question 18 (4 marks)

Alex and Bobby and a group of students were camping at a site. Alex walked with some students on a bearing of 037° for 9 km to location A. Bobby, with the rest of the students, walked on a bearing of 102° for 13 km to location B.



NOT TO
SCALE

- (a) Show that the angle AOB is 65° .

1

$$\begin{aligned}\angle AOB &= \angle NOB - \angle NOA \\ &= 102^\circ - 37^\circ \\ &= 65^\circ \quad (1)\end{aligned}$$

- (b) Find the distance AB in km, correct to 2 decimal places.

1

$$\begin{aligned}AB^2 &= 9^2 + 13^2 - 2 \times 9 \times 13 \times \cos 65 \\ &= 381.6142 \dots \\ AB &= \sqrt{381.6142 \dots} \quad (1) \\ &= 19.52 \dots = 19.52 \text{ km (2 d.p.)}\end{aligned}$$

- (c) Find the bearing of Bobby's group from Alex's group.
Give your answer correct to the nearest degree.

2

$$\begin{aligned}\frac{\sin \theta}{13} &= \frac{\sin 65}{12.29} \\ \theta &= \sin^{-1} \left(\frac{13 \times \sin 65}{12.29} \right) \\ &= 73.42\end{aligned}$$

$$\begin{aligned}\text{Bearing} &= 360 - 143 - 73.42 \\ &= 143.58 \\ &= 144^\circ\end{aligned}$$

Working toward θ using
sine or cosine rule

correct bearing

Question 19 (4 marks)

Solve the following equations:

(a) $\cos \theta = -\frac{1}{\sqrt{2}}, \quad 0 < \theta < 360^\circ$

1

$\theta = 135^\circ, 225^\circ$ (1)

(b) $2\sin^2 \theta - 3\sin \theta = -1, \quad 0 < \theta < 2\pi$

3

$2\sin^2 \theta - 3\sin \theta + 1 = 0$ let $u = \sin \theta$

$2u^2 - 3u + 1 = 0$

$(2u-1)(u-1) = 0$

$\therefore u = \frac{1}{2} \quad u = 1$

$\therefore \sin \theta = \frac{1}{2} \quad \sin \theta = 1$

$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{\pi}{2}$

[1] • let $u = \sin \theta$ + attempts to make quadratic
• attempt to factorise

[2] correct fact
 $u = \frac{1}{2} \quad u = 1$ (either
 $\sin \theta = \frac{1}{2} \quad \sin \theta = 1$ (either

Question 20 (2 marks)

Find the values of α and x such that:

$\alpha \sin x = 5 \sin 20^\circ + 3 \cos 70^\circ$

$\sin \theta = \cos(90^\circ - \theta)$

$\alpha \sin x = 5 \sin 20 + 3 \sin 20$
 $= 8 \sin 20$

$\therefore \alpha = 8, \quad x = 20^\circ$

[1] • using complementary identity ($\cos 70 = \sin 20$)
• attempt to add like terms
• either $x = 20^\circ$ or $\alpha = 8$

[2] correct values of α and x .

Question 21 (3 marks)

Prove that $\frac{1}{\cos \theta} + \cos \theta = (2 - \sin^2 \theta) \sec \theta$.

3

$$\begin{aligned} \text{LHS} &= \frac{1}{\cos \theta} + \cos \theta \\ &= \frac{1 + \cos^2 \theta}{\cos \theta} \\ &= \frac{1 + (1 - \sin^2 \theta)}{\cos \theta} \end{aligned} \quad \begin{aligned} &= \frac{2 - \sin^2 \theta}{\cos \theta} \\ &= (2 - \sin^2 \theta) \sec \theta \\ &= \text{RHS} \end{aligned}$$

End of Assessment

$$\begin{aligned} \text{① } \sin^2 \theta &= 1 - \cos^2 \theta \\ \cos^2 \theta &= 1 - \sin^2 \theta \\ \sec \theta &= \frac{1}{\cos \theta} \\ \frac{1}{\cos \theta} &= \sec \theta \end{aligned}$$

② LHS - comm. den
works towards
simplifying LHS

→ RHS

③ correct soln
correct s. out

④ if transposes
any terms.
or doesn't use
LHS/RHS.